

Ticklab — Investor One-Pager

> **Honest measurement for concentrated-liquidity LP strategies.**
> Backtest and risk-analytics infrastructure for Uniswap V3 and V4, built on
> real on-chain data with zero API keys required.

Problem

Every concentrated-liquidity LP needs to know whether a backtest is honest before they commit capital. Today there are two options:

- **Black-box dashboards** (APY.vision, Revert, DexScreener) — look polished but hide the methodology, can't be re-run, and routinely project 30-day returns that are 5–10× the realized number once you account for impermanent loss and out-of-range time.
- **Hand-rolled notebooks** — accurate but slow; every quant team rebuilds the same V3 in-range math from scratch, and most of them ship a buggy version because the math is subtle (tick spacing, feeShare at retail size, sqrtPriceX96 rounding).

The cost of a wrong backtest is not a number — it's a six-figure LP position that went out of range during a volatility event and ate the entire fee revenue plus principal.

Solution

Ticklab is an open-source, reproducible simulator + risk-analytics platform for V3 and V4. Three things make it different:

1. **Empirical validation, not vibes.** A reproducible north-star harness compares the simulator's output against a ground-truth LP P&L replay on real DeFi Llama + Binance OHLC data. The current median absolute error is **1.55 percentage points** on 30-day cumulative returns across **15 V3 pools** — a **6× improvement** over the previous baseline. Every CSV lives in the repo under ``validation-results/``, and a live ``/validation`` page renders the latest run.
2. **Zero API keys by default.** All data sources (DeFi Llama, Binance OHLC, public RPCs) are free and open. The one optional key — Covalent GoldRush for per-swap historical data — is env-gated and disabled out of the box.
3. **Production engineering, not a notebook.** Standalone SDK (``@ticklab/sdk``, ESM, zero deps), 33 REST API routes, OpenAPI 3.1 spec, Prometheus ``/api/metrics``, containerized Alpine image (~85 MB, non-root), one-click Vercel deploy, CI on three workflows, Playwright E2E suite.

Why now

- Uniswap V3 still holds **~\$2.5B in TVL** across 5+ chains with no credible open-source backtester that has empirical accuracy evidence attached to it.
- V4 launches are ramping through 2026, and **every V4 hook strategy** (dynamic fees, MEV capture, yield layers) needs a simulator that can model them. V4 is already shipped in Ticklab.
- Regulators and auditors are starting to ask funds "where did this backtest number come from?" — the answer needs to be a Git URL with a hash, not a screenshot of a vendor dashboard.

Traction (open-source signals, last 30 days)

Metric	Value
Root test suite	**288 / 288 green**
SDK test suite	**35 / 35 green**
Playwright E2E	**7 / 8 specs**
North-star median abs error	**1.55 pp** (was 7.91)
Coverage (V3 pools)	**15 / 18** free, **18 / 18** with Covalent
API routes	33
Marketing assets shipped	Hero clips × 4, explainers × 4, infographics × 2, whitepaper, 4 tweet threads, 2 articles, one-pager
Documentation surface	README, CONTRIBUTING, SECURITY, COC, NORTH_STAR_REPORT, CHANGELOG, /validation

Market

Segment	Size estimate
Active V3 LPs (Ethereum + L2s)	~50,000 wallets with > \$1k positions
Quant funds running LP strategies	~120 globally
DeFi desks, treasury teams	~500 globally
Per-seat tooling spend	\$200–\$2,000 / month

The wedge is "honest backtest that anyone can verify" – once that's established, the monetizable surface is: hosted tier with custom SLM sources (Coin Metrics, Glassnode), strategy-distribution marketplace (funds list strategies, LPs subscribe), and an audited-tier for institutional LPs.

What's next (90 days)

1. **Coin Metrics + Glassnode adapters** (paid, env-gated, opt-in) – fills the last 10% of measurement gaps and gives paid users a reason to upgrade.
2. **Backtest confidence interval** – every projection returns a confidence band based on the historical match between sim and ground truth for that pool. The math for this is already in ``lib/analytics/confidence.ts``; the UI is the next step.
3. **V4 hook marketplace** – protocols list their hooks (dynamic fee, MEV capture, etc.) with audited backtests against real volume. LPs browse + subscribe.
4. **Hosted tier** – same SDK, plus SLM data, plus the audit trail that funds need for compliance.

Team

Solo founder/engineer. Background building LP infrastructure for small funds (DeFi desk tooling) and contributing to public DeFi data sources. The north-star harness methodology and the median abs error reduction are documented in [``NORTH_STAR_REPORT.md``] (`../NORTH_STAR_REPORT.md`).

Ask

- **Strategic intros** to quant fund CTOs, DeFi-desk heads, and treasury teams running concentrated-LP strategies. Email ``founders@ticklab.dev``.
- **Data partnerships** with paid DeFi data providers (Coin Metrics, Glassnode, Dune) for the hosted tier.
- **Capital** – pre-seed, \$400–800k, 18-month runway, for the hosted tier + first two FT hires (frontend + data engineer). Use of funds: 50% engineering, 25% data partnerships, 25% GTM.

Links

- Live demo: ``https://ticklab.vercel.app``
- Repo: ``https://github.com/Mine-FNL/ticklab``
- Validation page: ``ticklab.vercel.app/validation``
- Whitepaper: ``marketing/whitepaper/ticklab-whitepaper.md``
- One-pager (cold email variant): ``marketing/one-pager/ticklab-one-pager.pdf``

*This document is rendered from ``marketing/one-pager/investor-one-pager.md``.
The numbers above are reproducible: run ``npm run validate:northstar`` in
the repo and inspect ``validation-results/validation-<timestamp>.csv``.*